Hyperspectral Image Denoising Based on Hyper-Laplacian Total Variation in Spectral Gradient Domain

Fang Yang, Qiangfu Hu, and Xin Su

***Abstract*— This article introduces a novel hyperspectral image (HSI) denoising method based on Hyper-Laplacian Total Variation in Spectral Gradient (HLTVSG). Our research identifies that compared to the original HSI, the spectral gradient of HSIs exhibits both a lower rank and a more favorable structure for total variation (TV) regularization due to its lower image entropy. To capitalize on the low-rank property of HSIs, we employ a subspace representation approach within the spectral gradient domain. Beyond low rankness, HSIs also possess abundant spatial information, which is often regularized by using TV. Traditional TV inherently assumes that the ℓ_1-norm of spatial gradient follows the Laplacian distribution. However, our analysis reveals that the spatial gradient of the subspace representation coefficients obeys a hyper-Laplacian distribution. Therefore, we introduce the hyper-Laplacian TV (HLTV) to better constrain the spatial local smoothness (LSS) of the representation coefficients. In addition, to eliminate the sparse noise and the stripe noise, an ℓ_1-norm and low-rank constraint are also employed. Finally, an augmented Lagrange multiplier (ALM) algorithm is utilized to solve the proposed model. Extensive experimental results on simulation and real datasets indicate that the proposed method outperforms many state-of-the-art methods both visually and quantitatively.**

***Index Terms*— Hyper-Laplacian total variation (HLTV), hyperspectral image (HSI), spectral gradient domain, subspace representation.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture diverse imaging effects of a real scene across hundreds of continuous spectral bands, providing an abundance of rich spectral and spatial information. They play an important role in a wide range of tasks, including environmental monitoring [1], object detection [2], change detection [3], [4], precision agriculture [5], urban analysis [6], military applications [7], and so on. High-quality HSIs can significantly enhance the performance of these tasks [8], [9], [10]. However, HSIs are inevitably interfered with by various noises during acquisition, such as Gaussian noise, impulse noise, dead lines, and stripes. These mixed noises severely degrade the quality of HSI, hindering the performance of HSI in subsequent applications. Therefore, how to remove the noise from HSI effectively becomes a critical issue.

A simple idea is to independently treat each band as a gray-scale image and directly apply classical gray-scale image restoration methods, such as nonlocal means (NLMs) [11], block-matching 3-D filtering (BM3D) [12], and weighted nuclear norm minimization (WNNM) [13]. A more logical way is to treat HSI as a 3-D data cube for processing, such as BM4D [14] and VBM4D [15]. However, due to the absence of intrinsic prior constraints, these methods often yield unsatisfactory results when handling images contaminated by mixed noise. To address these challenges, a growing body of HSI denoising research that integrates both spectral and spatial information has been proposed. These methods can be generally categorized into two main groups: model-based and deep learning-based. Model-based methods formulate HSI denoising as a constrained optimization problem, explicitly leveraging prior knowledge of HSIs, such as low-rank properties, local smoothness (LSS), and nonlocal self-similarity (NSS). In contrast, deep learning-based methods aim to directly learn the mapping function between noisy and clean HSIs from paired training datasets, thus avoiding the need for explicit prior modeling.

To deal with complex mixed noises, the robust principal component analysis (RPCA) framework [16] has been introduced and widely used for HSI noise removal. Inspired by RPCA, Zhang et al. [17] first proposed a framework for low-rank matrix recovery (LRMR) combined with sparsity. LRMR successfully exploits the low-rank property of clean HSIs for denoising but fails to preserve spatial information in complex noise, resulting in unsatisfactory performance. TV regularization is able to preserve spatial LSS of images [18] and has been widely used in HSI denoising. For instance, He et al. [19] introduced a TV-based low-rank matrix restoration approach (LRTV). This method employs bandwise TV regularization to preserve spatial LSS, thereby enhancing the denoising performance. Furthermore, to improve the performance of bandwise TV regularization, spatial–spectral TV (SSTV) and NSS regularizations have been extensively applied in HSI denoising [20], [21]. However, TV may not be an ideal regularizer for HSI denoising. Most HSI denoising methods that rely on LSS priors utilize the TV regularization

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with ℓ_1-norm. The regularization prior of the ℓ_1-norm follows a Laplacian distribution, which fails to accurately capture the underlying distribution of gradient data and often results in a heavy-tailed phenomenon.

Besides, the aforementioned low-rank-based methods inevitably disrupt spatial structures because they convert the HSI cube into a 2-D matrix for low-rank modeling. To maintain the intrinsic spatial structure of an HSI, many methods propose treating it as a 3-D tensor. Different tensor decompositions have been proposed, including tensor singular value decomposition (SVD) [22], [23], Tucker decomposition [24], [25], CANDECOMP/PARAFAC (CP) decomposition [26], and tensor ring decomposition [22]. In addition, tensor-based methods are often combined with other priors to achieve better results in removing complex noise. For example, Wang et al. [24] employed Tucker decomposition and incorporated SSTV to remove the mixed noise (LRTDTV). Xue et al. [26] proposed a nonlocal low-rank regularization CP tensor decomposition model. Wang et al. [27] built a unique tensor-correlated TV (CTV) regularizer, which essentially encodes both low rankness and LSS priors of a tensor simultaneously. Although tensor-based techniques achieve effective denoising while maintaining the intrinsic structure, they involve complex computations and higher data dimensionality, resulting in increased computational costs.

Recently, the subspace representation-based methods have gained attention for reducing HSI denoising complexity, as each spectral feature can be represented as a linear combination of a few pure spectral endmembers [28]. Thus, suppose that the dimension of the spectral subspace is r. A clean HSI $\mathcal{X} \in \mathbb{R}^{M\times N\times P}$ can be represented as $\mathcal{X} = \mathcal{U} \times_3 \mathbf{V}$, where $\mathbf{V} \in \mathbb{R}^{P\times r}(r \ll P)$ is the spectral basis matrix and $\mathcal{U} \in \mathbb{R}^{M\times N\times r}$ represents the spatial representation coefficients (SRCs) [29]. This model captures global spectral low rankness, avoiding the need for large-scale SVD computations. Taking advantage of this model, Zhuang and Bioucas-Dias [30] fixed the learned spectral basis and used the well-known BM3D denoising algorithm to filter SRCs. He et al. [31] proposed an ensemble paradigm based on subspace representation and optimized the denoising results through iteration. Chen et al. [32] propose a fast large-scale HSI denoising method based on noniterative low-rank subspace representation. Zheng et al. [33] not only considered the spatial LSS inherent in the HSI subspace but also capitalized on the continuity within the spectral domain (LRTF-DFR). Based on LSS prior of $\mathcal{U}$, Peng et al. [34] integrated TV regularization within the subspace representation framework (RCTV), balancing denoising performance and speed.

In addition, some HSI denoising methods have been proposed based on the spectral gradient domain. Compared to the original HSI, the spectral gradient domain exhibits a lower rank and reduced image entropy, facilitating more efficient mixed noise removal. For example, Sun et al. [35] proposed an algorithm that exploits the low rankness of the spectral gradient domain to reduce mixed noise in HSI. They further developed a method incorporating more prior knowledge based on spectral difference-induced TV and low-rank approximation [36]. Yang et al. [37] projected the HSI data into the spectral gradient domain using spectral difference transformation and applied graph-based denoising. Liu et al. [38] proposed a novel tensor-based HSI restoration model with low-rank modeling in the gradient domain within a unified tensor representation framework. Although these spectral gradient domain-based methods are feasible in practice, the related spectral difference minimization problem involves an additional step that recovers the clean HSI in spectral gradient domain [35], [38], [39], leading to an increase in computational cost. Therefore, effectively leveraging the advantages of the spectral gradient domain to enhance restoration performance while reducing costs is a critical issue.

Aside from model-based restoration methods, deep learning-based approaches have gained significant attention for HSI denoising. These methods typically rely on training a supervised deep neural network (DNN) with paired clean-noisy HSI data [40], [41], [42], [43]. Notable examples include the following: Yuan and Ng [40] proposed a deep residual convolutional neural network (CNN) incorporating multiscale and multilevel feature representations to effectively address HSI denoising. Zhuang and Ng [41] introduced FastHyMix, a parameter-free method for HSI mixed noise removal. Fu et al. [44] developed the spatial–spectral recurrent transformer U-Net (SSRT-UNet), which integrates low-rank and NSS properties within a unified SSRT block. However, supervised learning methods are fundamentally limited by their reliance on large amounts of paired training data, which is often scarce or difficult to obtain in practical scenarios. Furthermore, these models typically assume a specific noise prior, leading to poor generalization when confronted with varying noise characteristics. Such constraints limit the adaptability of supervised methods, particularly in real-world applications where noise types may not be well-defined or may vary significantly across datasets.

In contrast, unsupervised learning methods have emerged as an alternative by utilizing the inherent structure and characteristics of HSI data to improve generalization, thereby eliminating the need for paired training data. For example, Sidorov and Hardeberg [45] extended the deep image prior (DIP) technique to the HSI domain, a method that does not require any labeled data but rather learns an image prior during the optimization process. Building on this, Shi et al. [46] proposed a double subspace deep prior model, which further enhances the unsupervised DIP approach by exploiting subspace learning to better capture HSI-specific structures. While unsupervised learning methods alleviate the need for paired data and exhibit better generalization to unseen noise types, they often suffer from several drawbacks. One significant limitation is the need for extensive optimization, as unsupervised methods typically require a large number of iterations to converge to an optimal solution. Moreover, these methods frequently face interpretability challenges, as the learned priors may not always align with human-understandable features or structures.

Based on the above analysis, we propose a novel low-rank smoothness-based denoising method in the spectral gradient domain, referred to as Hyper-Laplacian Total Variation in Spectral Gradient (HLTVSG), which enhances denoising

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performance. Inspired by the low rankness of spectral gradient, we introduce a subspace representation in the spectral gradient domain, i.e., $D_3(\mathcal{X}) = \mathcal{U} \times_3 \mathbf{V}$, where $D_3(\cdot)$ represents the first-order difference operator along the spectral mode. The SRCs $\mathcal{U}$ can effectively preserve the spatial information inherent in $D_3(\mathcal{X})$. In addition, from a statistical view, we found that the gradient of the SRCs follows the hyper-Laplacian distribution. Hence, we introduce a hyper-Laplacian TV (HLTV) to preserve the LSS of HSIs instead of using TV. The major contributions of this article are given as follows.

1) *Reformulating HSI Denoising in the Spectral Gradient Domain for Enhanced Low-Rank and Smoothness Modeling:* This work proposes a novel denoising framework by exploiting the spectral gradient domain, which inherently exhibits stronger low-rank property and lower entropy compared to the original HSI domain. The low-rank property in the spectral gradient domain allows for a more efficient representation of spectral correlations. We capture this property through subspace representation, avoiding the computational complexity of large-scale SVD while maintaining high accuracy. In addition, the low entropy provides a better foundation for TV regularization, improving the modeling of spatial smoothness and noise suppression.
2) *Enhancing Spatial LSS With HLTV:* Our observations indicate that compared to traditional TV, HLTV more effectively captures the spatial gradient distribution of the SRCs. This allows HLTV to better preserve fine spatial structures, including edges and subtle texture details, while suppressing noise.
3) *Effective Algorithm and Superior Performance:* The proposed model is solved using the ALM algorithm, ensuring efficient and reliable optimization. Extensive experiments on both simulated and real datasets demonstrate that the proposed method outperforms state-of-the-art HSI denoising methods.

The rest of this article is organized as follows. Section II introduces some notations and provides preliminary knowledge. Section II-C presents our proposed optimization model based on subspace representation in spectral gradient domain and its corresponding optimization procedure. Section IV conducts extensive experiments to demonstrate the superiority of our proposed model. Finally, Section V summarizes this article.

II. Notations and Preliminaries

A. Notations

Throughout this article, tensors are denoted by capitalized calligraphic letters, i.e., $\mathcal{X}$. Matrices are denoted by boldface capital letters, i.e., $\mathbf{X}$. Vectors are represented as lowercase boldface letters, i.e., $\mathbf{x}$, and scalars are denoted with lowercase letters or capital letters, i.e., x or X. For an N-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$, $x_{i_1,i_2,\ldots,i_N}$ is its element located at $(i_1, i_2, \ldots, i_N)$. Then, the inner product of two tensors $\mathcal{X}$ and $\mathcal{Y}$ of the same size is defined as $\langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{i_1,i_2,\ldots,i_N} x_{i_1,i_2,\ldots,i_N} \cdot y_{i_1,i_2,\ldots,i_N}$. Moreover, the Frobenius norm of tensor $\mathcal{X}$ is given by $\|\mathcal{X}\|_F = (\langle \mathcal{X}, \mathcal{X} \rangle)^{1/2}$. The ℓ_1 norm is defined as $\|\mathcal{X}\|_1 = \sum_{i_1,i_2,\ldots,i_N} |x_{i_1,i_2,\ldots,i_N}|$, and the ℓ_q norm is defined as $\|\mathcal{X}\|_q = (\sum_{i_1,i_2,\ldots,i_N} |x_{i_1,i_2,\ldots,i_N}|^q)^{1/q}$. The mode-$n$ unfolding of N-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ can be categorized into horizontal and vertical unfoldings. Specifically, the mode-n horizontal unfolding is expressed as $\mathbf{X}_{(n)} \in \mathbb{R}^{I_n \times (I_1 \cdots I_{n-1} I_{n+1} \cdots I_N)}$, while the mode-$n$ vertical unfolding is denoted as $\mathbf{X}^{(n)} \in \mathbb{R}^{(I_1 \cdots I_{n-1} I_{n+1} \cdots I_N) \times I_n}$. Furthermore, the mode-$n$ multiplication of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ with a matrix $\mathbf{A} \in \mathbb{R}^{J \times I_n}$ is represented as $\mathcal{Y} = \mathcal{X} \times_n \mathbf{A}$, where $\mathcal{Y} \in \mathbb{R}^{I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$ and $y_{i_1,\ldots,i_{n-1},j,i_{n+1},\ldots,i_N} = \sum_{i_n} x_{i_1,\ldots,i_N} \cdot \mathbf{A}_{j,i_n}$.

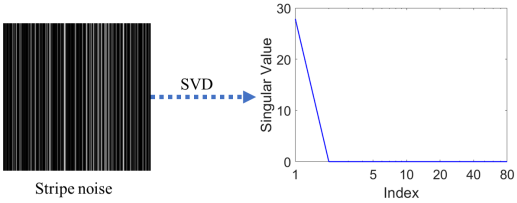


Fig. 1. Low-rank property of the stripe noise.

B. Low Rankness of Stripe Noise

Stripe noise exhibits distinct structural and directional characteristics. To model the stripes appropriately, Chang et al. [47] conducted a quantitative analysis of the properties of stripes in HSI and found that stripe noise can be effectively modeled using bandwise low-rank constraints. SVD was applied to decompose the simulated stripe noise image, as shown in Fig. 1. It can be observed that the singular values of the stripe noise rapidly decay to zero, demonstrating that the stripe noise has a low-rank property. Therefore, the stripe noise $\mathcal{S}$ of an HSI can be well modeled by the following low-rank regularization:

$$\sum_{b=1}^{P} \operatorname{rank}(\mathbf{S}_b), \quad \text{s.t.} \operatorname{rank}(\mathbf{S}_b) \le r_s \tag{1}$$

where $\mathbf{S}_b$ represents the stripe noise at band b and r_s is the upper bound of the rank of $\mathbf{S}_b$.

Note that while both the clean HSI and stripe noise exhibit low-rank properties, these characteristics arise from distinct mechanisms.

1) The clean HSI derives its low-rank structure from global spectral correlations across spectral bands.
2) Stripe noise, conversely, exhibits local spatial low rankness within individual spectral bands due to repetitive columnar patterns.

Crucially, the low-rank property of the clean HSI operates in the spectral domain, whereas stripe noise's low rankness resides purely in the spatial domain. This orthogonality ensures that bandwise HSI data, which lack inherent low rankness in isolation, do not interfere with stripe noise regularization. When applying the constraint [see (1)], the spatial-domain low-rank modeling of stripes remains decoupled from the spectral-domain characteristics of the clean HSI.

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C. HSI Degradation Model

Given a noisy HSI $\mathcal{Y} \in \mathbb{R}^{M\times N\times P}$, where M and N are spatial sizes and P is the spectral number, this article considers a mixture of three different types of noises, including Gaussian noise, sparse noise, and stripe noise. Therefore, the noisy HSI $\mathcal{Y}$ can be described by the following model:

$$\mathcal{Y} = \mathcal{X} + \mathcal{B} + \mathcal{S} + \mathcal{N} \tag{2}$$

where $\mathcal{X}$ denotes the clean HSI, $\mathcal{B}$ is the sparse noise, $\mathcal{S}$ is the stripe noise, and $\mathcal{N}$ is the Gaussian noise. Given the observed noisy data $\mathcal{Y}$, we aim to recover the clean HSI $\mathcal{X}$ from it.

The problem of recovering $\mathcal{X}$ from (2) is a challenging ill-posed problem. To address the problem, the prior regularization of different unknown variables is required. Therefore, the optimization model of the mixed noise removal problem can be formulated as

$$\begin{aligned} &\arg\min_{\mathcal{X},\mathcal{S},\mathcal{B}} J_1(\mathcal{X}) + \lambda J_2(\mathcal{B}) + \beta J_3(\mathcal{S}) \\ &\quad \text{s.t. } \|\mathcal{Y} - \mathcal{X} - \mathcal{S} - \mathcal{B}\|_F^2 \le \varepsilon \end{aligned} \tag{3}$$

where $J_1(\mathcal{X})$, $J_2(\mathcal{B})$, and $J_3(\mathcal{S})$ are the regularization terms that describe the prior information of the clean HSI, sparse noise, and stripe noise, respectively. λ and β are nonnegative regularization parameters that control the tradeoff between each regularization term. Generally, $J_2(\mathcal{B})$ is a sparse regularization term [17], which can be expressed as $J_2(\mathcal{B}) = \|\mathcal{B}\|_1$. In addition, as analyzed in Section II-B, the stripe regularization is represented as $J_3(\mathcal{S}) = \sum_{b=1}^{P} \|\mathbf{S}_b\|_*$.

III. Proposed Method

A. Spectral Gradient Domain

The spectral gradient domain is the space where spectral gradients of HSIs are represented and analyzed. In this domain, spectral information can be obtained by applying a difference operator across adjacent spectral bands. This representation captures subtle spectral variations and enhances critical features essential for tasks such as denoising. By leveraging the inherent low-rank properties of spectral data and spatial correlations within the image, the spectral gradient domain enables more effective and efficient processing.

1) Low-Rank Property: Since the spectral difference operator $D_3(\cdot)$ is linear, the spectral gradient domain also inherits a low-rank property. Fig. 2 illustrates the singular values of both the original HSI and the spectral gradient of HSI by vectorizing each band into a vector, transforming the 3-D tensor into 2-D matrix. The singular values are denoted by the blue and red curves, respectively. We can see that the red curves converge to zero at a smaller index than the blue curves, demonstrating that the spectral gradient domain not only possesses low-rank property but also exhibits a lower rank than the original domain.

The lower rank property in the spectral gradient domain makes it particularly well-suited for subspace representation-based denoising methods. This is due to that the subspace representation generally involves projecting data onto a low-dimensional space, allowing for the efficient capture of the intrinsic structure of the data while minimizing the influence

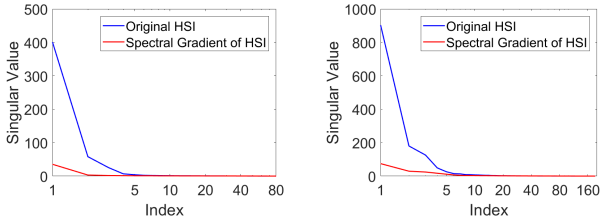


Fig. 2. Spectral low-rank prior of the spectral gradient of HSI. (a) Pavia City Center. (b) Washington DC Mall.

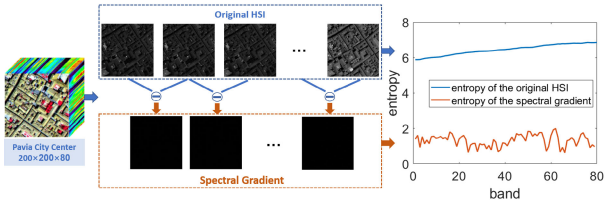


Fig. 3. Image entropy of each band of the original HSI and its spectral gradient.

of noise and redundant information. In the spectral gradient domain, the inherently lower rank of the data enables a more compact representation within a low-dimensional subspace, thereby enhancing the efficiency of both dimensionality reduction and denoising processes.

2) Low Image Entropy: In this article, we employ TV-based regularization to preserve the spatial information of HSIs. TV is particularly effective for images with strong local similarity, where pixel values exhibit smooth, gradual variations across neighboring regions. This strong local similarity often corresponds to low entropy, indicating a lack of abrupt intensity variations and a more predictable, homogeneous structure. Hence, low entropy reflects uniform intensity distributions, creating ideal conditions for TV denoising.

The original clean HSI contains substantial information and variance, resulting in relatively high image entropy. However, HSIs often exhibit significant redundancy among spectral bands, meaning that adjacent bands are highly correlated and contain overlapping information. Consequently, the spectral gradient, which measures the difference between adjacent bands, has reduced information content and lower entropy compared to the original HSI. This reduction in entropy makes the spectral gradient more suitable for TV denoising. Fig. 3 illustrates the various bands of the original HSI and its spectral gradient, along with their entropy values, respectively. Because of the similarity between bands, the spectral gradient contains less redundant information. Notably, the entropy of the original HSI is significantly higher than that of the spectral gradient, further indicating that the spectral gradient is better suited for TV denoising.

B. HLTV Regularizer

HLTV is an advanced regularization technique designed for image denoising, particularly in the presence of complex noise. Unlike traditional TV, which assumes that gradients follow a Laplacian distribution that poorly fits both the mean and the

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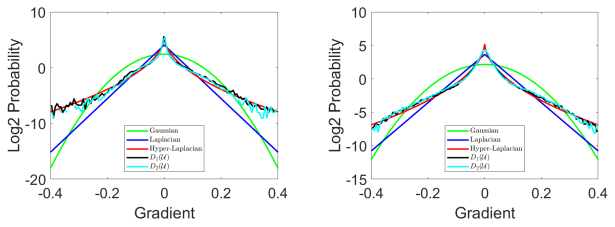


Fig. 4. Empirical fitting curves of image gradients for SRCs. (a) Pavia City Center. (b) Washington DC Mall.

tail, HLTV uses a hyper-Laplacian distribution with heavier tails, making it more effective for handling real-world noise commonly found in HSIs [48], [49], [50].

As discussed above, HSIs are characterized by strong spectral correlations, which allow their spectral gradients to be represented as $D_3(\mathcal{X}) = \mathcal{U} \times_3 \mathbf{V}$, where $\mathcal{U}$ preserves spatial information. With an orthogonal endmember matrix $\mathbf{V}$, $\mathcal{U}$ effectively captures spatial prior information [34]. Applying HLTV to $\mathcal{U}$ can better model the gradient distribution, improving denoising performance by preserving both spectral and spatial details. As shown in Fig. 4, the hyper-Laplacian distribution has heavier tails and closely matches the gradient distribution of $\mathcal{U}$, outperforming both Gaussian and Laplacian models. Leveraging this prior knowledge, we employ the HLTV to characterize the gradient distribution in our method

$$\|\mathcal{U}\|_{\mathrm{HLTV}} = \sum_{i,j,k} |u_{i,j,k} - u_{i-1,j,k}|^{q_1} + \left|u_{i,j,k} - u_{i,j-1,k}\right|^{q_2} \quad (4)$$

where $u_{i,j,k}$ is the (i, j, k)th entry of $\mathcal{U}$. The values of power parameter q_1 and q_2 $(0 \le q_1, q_2 \le 1)$ are used to regulate the sparsity of the hyper-Laplacian. The HLTV is also equivalently expressed as

$$\|\mathcal{U}\|_{\mathrm{HLTV}} = \|D_1(\mathcal{U})\|_{q_1}^{q_1} + \|D_2(\mathcal{U})\|_{q_2}^{q_2} \quad (5)$$

where $D_1(\cdot)$ and $D_2(\cdot)$ are linear operators corresponding to the first-order discrete differences in the horizontal and vertical directions, respectively.

C. Proposed Model

In this work, we leverage the low-rank property and smoothness of HSIs in the spectral gradient domain to develop our optimization model. The flowchart of the proposed HLTVSG method is shown in Fig. 5. First, we apply spectral difference transformation to map the HSI into the spectral gradient domain. Next, a low-rank subspace representation is used to capture spectral correlations. Subsequently, the obtained SRCs are denoised using HLTV regularization. Finally, an iterative refinement process is applied to achieve the final denoised result. The detailed formulation of the restoration model HLTVSG is described as follows:

$$\begin{aligned} \min_{\mathcal{X},\mathcal{B},\mathcal{S},\mathcal{U},\mathbf{V}} & \sum_{i=1}^{2} \|D_i(\mathcal{U})\|_{q_i}^{q_i} + \lambda\|\mathcal{B}\|_1 + \beta \sum_{b=1}^{P} \|\mathbf{S}_b\|_* \\ \text{s.t.} \quad & \|\mathcal{Y} - \mathcal{X} - \mathcal{S} - \mathcal{B}\|_F^2 \le \xi, \quad \mathrm{rank}(\mathbf{S_b}) \le r_s \\ & D_3(\mathcal{X}) = \mathcal{U} \times_3 \mathbf{V}, \quad \mathbf{V}^{\mathrm{T}}\mathbf{V} = \mathbf{I} \end{aligned} \quad (6)$$

where λ and β are the parameters that control the sparse regularizer for sparse noise and the low-rank regularizer for stripe noise, respectively. In addition, r_s represents the upper bound of the stripe matrix. The proposed restoration model aims to optimize the variables $\mathcal{X}$, $\mathcal{B}$, $\mathcal{S}$, $\mathcal{U}$, and $\mathbf{V}$ simultaneously, which can be solved using an alternating minimization strategy. Next, we will introduce the optimization process and solve the optimization problem based on the ALM method.

D. Optimization Procedure

As previously mentioned, the ALM method is used to solve the minimization of (6) by introducing auxiliary variables $\mathcal{Z}_1$ and $\mathcal{Z}_2$. Therefore, the optimization problem is equivalent to the following formulation:

$$\begin{aligned} \min_{\mathcal{X},\mathcal{B},\mathcal{S},\mathcal{U},\mathbf{V},\mathcal{Z}_i} & \sum_{i=1}^{2} \|\mathcal{Z}_i\|_{q_i}^{q_i} + \lambda\|\mathcal{B}\|_1 + \beta \sum_{b=1}^{P} \|\mathbf{S_b}\|_* \\ \text{s.t.} \quad & \|\mathcal{Y} - \mathcal{X} - \mathcal{S} - \mathcal{B}\|_F^2 \le \xi, \quad \mathrm{rank}(\mathbf{S_b}) \le r_s \\ & D_3(\mathcal{X}) = \mathcal{U} \times_3 \mathbf{V}, \quad \mathbf{V}^{\mathrm{T}}\mathbf{V} = \mathbf{I} \\ & D_i(\mathcal{U}) = \mathcal{Z}_i, \quad i = 1, 2. \end{aligned} \quad (7)$$

Therefore, the augmented Lagrangian function is expressed as

$$\begin{aligned} & \mathcal{L}\left(\mathcal{X}, \mathcal{B}, \mathcal{S}, \mathcal{U}, \mathbf{V}, \{\mathcal{Z}_i\}_{i=1}^2, \{\Gamma_i\}_{i=1}^4\right) \\ & = \sum_{i=1}^{2} \left\|\mathcal{Z}_i\right\|_{q_i}^{q_i} + \lambda\left\|\mathcal{B}\right\|_1 + \beta \sum_{b=1}^{P} \left\|\mathbf{S}_b\right\|_* \\ & \quad + \frac{\mu}{2}\|D_3(\mathcal{X}) - \mathcal{U} \times_3 \mathbf{V}\|_F^2 + \langle \Gamma_3, D_3(\mathcal{X}) - \mathcal{U} \times_3 \mathbf{V}\rangle \\ & \quad + \frac{\mu}{2} \sum_{i=1}^{2} \|D_i(\mathcal{U}) - \mathcal{Z}_i\|_F^2 + \sum_{i=1}^{2} \langle \Gamma_i, D_i(\mathcal{U}) - \mathcal{Z}_i\rangle \\ & \quad + \frac{\mu}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{B} - \mathcal{S}\|_F^2 + \langle \Gamma_4, \mathcal{Y} - \mathcal{X} - \mathcal{B} - \mathcal{S}\rangle \\ & \text{s.t.} \quad \mathrm{rank}(\mathbf{S_b}) \le r_s, \quad \mathbf{V}^{\mathrm{T}}\mathbf{V} = \mathbf{I} \end{aligned} \quad (8)$$

where μ represents the penalty parameter and $\{\Gamma_i\}_{i=1}^4$ denotes the Lagrange multipliers. According to the ALM method, we divide the problem (8) into six suboptimization problems and solve them by alternating iterations.

E. Optimization Process

1) Optimization Regarding $\{\mathcal{Z}_i\}_{i=1}^2$:

$$\mathcal{Z}_i = \arg\min_{\mathcal{Z}_i} \|\mathcal{Z}_i\|_{q_i}^{q_i} + \frac{\mu}{2}\left\|D_i(\mathcal{U}) - \mathcal{Z}_i + \frac{\Gamma_i}{\mu}\right\|_F^2. \quad (9)$$

According to [51], the nonconvex minimization problem in (9) can be efficiently solved using the generalized soft-thresholding (GST) algorithm.

2) Optimization Regarding $\mathcal{X}$:

$$\begin{aligned} \mathcal{X} = \arg\min_{\mathcal{X}} & \frac{\mu}{2}\left\|D_3(\mathcal{X}) - \mathcal{U} \times_3 \mathbf{V} + \frac{\Gamma_3}{\mu}\right\|_F^2 \\ & + \frac{\mu}{2}\left\|\mathcal{Y} - \mathcal{X} - \mathcal{B} - \mathcal{S} + \frac{\Gamma_4}{\mu}\right\|_F^2. \end{aligned} \quad (10)$$

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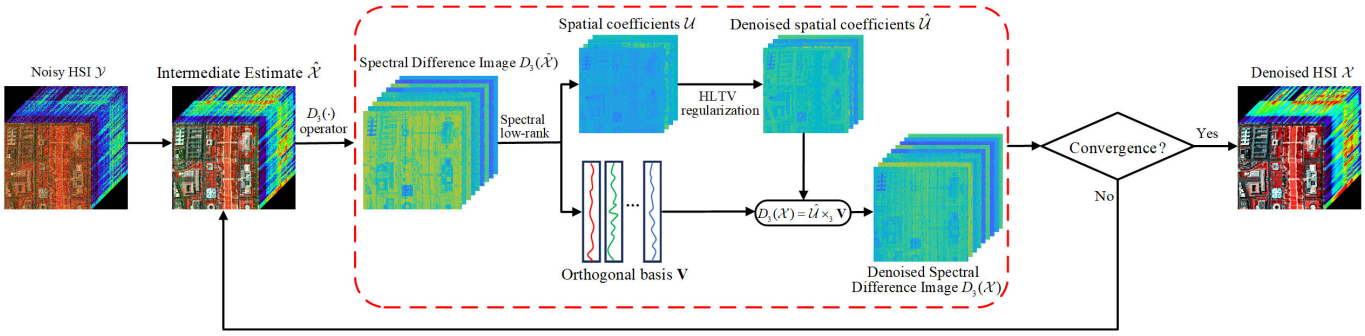


Fig. 5. Overview flowchart of the proposed HLTVSG method. First, the HSI data are projected into the spectral gradient domain via spectral difference transformation. Second, a low-rank subspace representation is applied to capture spectral correlations, representing the HSI as spectral basis and SRCs. Third, the HLTV regularization is applied to denoise the SRCs. Finally, iterative refinement is performed to obtain the final denoised result.

Optimizing this problem can be treated as solving the following linear system:

$$(\mu\mathbf{I}+\mu D_3^{\mathrm{T}}D_3)\mathcal{X} = \mu\left(\mathcal{Y}-\mathcal{B}-\mathcal{S}+\frac{\Gamma_4}{\mu}\right) + \mu D_3^{\mathrm{T}}\left(\mathcal{U}\times_3\mathbf{V}-\frac{\Gamma_3}{\mu}\right). \tag{11}$$

According to the fast Fourier transform (FFT) method, this problem can be effectively solved

$$\begin{cases} \mathcal{C}_1 = \mu\left(\mathcal{Y}-\mathcal{B}-\mathcal{S}+\dfrac{\Gamma_4}{\mu}\right)+\mu D_3^{\mathrm{T}}\left(\mathcal{U}\times_3\mathbf{V}-\dfrac{\Gamma_3}{\mu}\right) \\ \mathcal{X} = \mathcal{F}^{-1}\left[\dfrac{\mathcal{F}(\mathcal{C}_1)}{\mu\mathbf{1}+\mu|\mathcal{F}(D_3)|^2}\right] \end{cases} \tag{12}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ represent the FFT and inverse FFT, respectively, and $|\cdot|^2$ denotes the elementwise square.

3) Optimization Regarding $\mathcal{U}$:

$$\mathcal{U} = \arg\min_{\mathcal{U}} \frac{\mu}{2}\left\|D_3(\mathcal{X})-\mathcal{U}\times_3\mathbf{V}+\frac{\Gamma_3}{\mu}\right\|_F^2 + \frac{\mu}{2}\sum_{i=1}^{2}\left\|D_i(\mathcal{U})-\mathcal{Z}_i+\frac{\Gamma_i}{\mu}\right\|_F^2. \tag{13}$$

The optimization problem of variable $\mathcal{U}$ is similar to the optimization problem of variable $\mathcal{X}$ and equivalent to the following linear equation:

$$\left(\mu\mathbf{I}+\mu\sum_{i=1}^{2}D_i^{\mathrm{T}}D_i\right)\mathcal{U} = \mu\left(D_3(\mathcal{X})+\frac{\Gamma_3}{\mu}\right)\times_3\mathbf{V}^{\mathrm{T}} + \mu\sum_{i=1}^{2}D_i^{\mathrm{T}}\left(\mathcal{Z}_i-\frac{\Gamma_i}{\mu}\right). \tag{14}$$

Similarly, the optimization (14) can be solved via the FFT

$$\begin{cases} \mathcal{C}_2 = \mu\displaystyle\sum_{i=1}^{2}D_i^{\mathrm{T}}\left(\mathcal{Z}_i-\frac{\Gamma_i}{\mu}\right)+\mu\left(D_3(\mathcal{X})+\frac{\Gamma_3}{\mu}\right)\times_3\mathbf{V}^{\mathrm{T}} \\ \mathcal{U} = \mathcal{F}^{-1}\left[\dfrac{\mathcal{F}(\mathcal{C}_2)}{\mu\mathbf{1}+\mu\left(|\mathcal{F}(D_1)|^2+|\mathcal{F}(D_2)|^2\right)}\right]. \end{cases} \tag{15}$$

4) Optimization Regarding $\mathbf{V}$:

$$\mathbf{V} = \arg\min_{\mathbf{V},\mathbf{V}^{\mathrm{T}}\mathbf{V}=\mathbf{I}}\frac{\mu}{2}\left\|D_3(\mathcal{X})-\mathcal{U}\times_3\mathbf{V}+\frac{\Gamma_3}{\mu}\right\|_F^2. \tag{16}$$

According to [52], the solution of (16) can be obtained as

$$\begin{cases} \mathbf{M} = \left(D_3(\mathcal{X})+\dfrac{\Gamma_3}{\mu}\right)_{(3)}\mathcal{U}^{(3)} \\ \mathbf{C}\Sigma\mathbf{D}^{\mathrm{T}} = \mathrm{SVD}(\mathbf{M}) \\ \mathbf{V} = \mathbf{C}\mathbf{D}^{\mathrm{T}}. \end{cases} \tag{17}$$

5) Optimization Regarding $\mathcal{B}$:

$$\mathcal{B} = \arg\min_{\mathcal{B}}\lambda\|\mathcal{B}\|_1+\frac{\mu}{2}\left\|\mathcal{Y}-\mathcal{X}-\mathcal{B}-\mathcal{S}+\frac{\Gamma_4}{\mu}\right\|_F^2 = \mathrm{shrink_}\ell_1\left(\mathcal{Y}-\mathcal{X}-\mathcal{S}+\frac{\Gamma_4}{\mu},\frac{\lambda}{\mu}\right) \tag{18}$$

where shrink_ℓ_1 represents the soft threshold operation [53], which is defined as follows:

$$\mathrm{shrink_}\ell_1(x,\gamma) = \begin{cases} x-\gamma, & \text{if } x>\gamma \\ x+\gamma, & \text{if } x<\gamma \\ 0, & \text{else.} \end{cases} \tag{19}$$

6) Optimization Regarding $\mathcal{S}$:

$$\mathcal{S} = \arg\min_{\mathcal{S},\mathrm{rank}(\mathbf{S}_b)\le r_s}\beta\sum_{b=1}^{P}\|\mathbf{S}_b\|_*+\frac{\mu}{2}\left\|\mathcal{Y}-\mathcal{X}-\mathcal{B}-\mathcal{S}+\frac{\Gamma_4}{\mu}\right\|_F^2. \tag{20}$$

To solve problem (20), we can optimize the stripe noise matrix on each band separately. Then, we can rewrite (20) as follows:

$$\mathcal{S} = \arg\min_{\mathcal{S},\mathrm{rank}(\mathbf{S}_b)\le r_s} \times\sum_{b=1}^{P}\left(\beta\|\mathbf{S}_b\|_*+\frac{\mu}{2}\left\|\mathbf{S}_b-\left(\mathbf{Y}_b-\mathbf{X}_b-\mathbf{B}_b+\frac{\Gamma_{4,b}}{\mu}\right)\right\|_F^2\right). \tag{21}$$

The problem can be solved by the singular value shrinkage operation [54]. Therefore, the solution of stripe noise can be

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expressed as follows:

$$\begin{cases} \mathbf{C}\Sigma\mathbf{D}^{\mathrm{T}} = \mathrm{SVD}\left(\mathbf{Y}_b - \mathbf{X}_b - \mathbf{B}_b + \dfrac{\Gamma_{4,b}}{\mu}\right) \\ \mathrm{shrink_}\ell_*\left(\Sigma_r, \dfrac{\beta}{\mu}\right) = \mathrm{diag}\left\{\max\left(\Sigma_{\mathrm{rii}} - \dfrac{\beta}{\mu}, 0\right)\right\} \\ \Sigma_r = \mathrm{diag}(\sigma_i (1 < i < r)) \\ \mathbf{S}_b = \mathbf{C}\left(\mathrm{shrink_}\ell_*\left(\Sigma_{r_s}, \dfrac{\beta}{\mu}\right)\right)\mathbf{D}^{\mathrm{T}}. \end{cases} \tag{22}$$

The Lagrangian multipliers $\{\Gamma_i\}_{i=1}^4$ are updated in parallel

$$\begin{cases} \Gamma_i = \Gamma_i + \mu(D_i(\mathcal{U}) - \mathcal{Z}_i), \quad i = 1, 2 \\ \Gamma_3 = \Gamma_3 + \mu(D_3(\mathcal{X}) - \mathcal{U} \times_3 \mathbf{V}) \\ \Gamma_4 = \Gamma_4 + \mu(\mathcal{Y} - \mathcal{X} - \mathcal{B} - \mathcal{S}). \end{cases} \tag{23}$$

The procedure is summarized in Algorithm.

Algorithm 1 HLTVSG Algorithm

1: **Input:** Noisy HSI $\mathcal{Y}$, regularization parameters λ and β, stopping criterion ξ, maximum number of iterations $k_{\max}$, the upper bound rank r_s, and the rank r (the dimension of subspace).
2: **Initialize:** $\mathcal{X} = \mathcal{Y}$, $\mathcal{S} = \mathcal{B} = 0$, $\Gamma_1 = \Gamma_2 = 0$, $\Gamma_3 = \Gamma_4 = 0$, $\mathcal{U}$ and $\mathbf{V}$ are estimated by SVD, $\rho = 1.3$, $\mu = 0.1$ and $\mu_{\max} = 10^6$.
3: **for** k=1:$k_{\max}$ **do**
4: Update Z_i, $\mathcal{X}$, $\mathcal{U}$, $\mathbf{V}$, $\mathcal{B}$ and $\mathcal{S}$ via Eq. Eq: (10), Eq: (13), Eq: (16), Eq: (17), Eq: (19) and Eq: (23).
5: Update Γ_i for $i = 1, 2, 3, 4$ via Eq. Eq: (24).
6: Update the penalty parameter $\mu = \min(\rho\mu, \mu_{\max})$.
7: Check the convergence condition $\|\mathcal{X}^{k+1} - \mathcal{X}^k\|_F / \|\mathcal{X}^k\|_F \le \xi$.
8: **end for**
9: **Output:** Denoised HSI $\mathcal{X}$.

IV. Experimentation

To validate the superiority of the proposed method, we conduct extensive comparative experiments on both simulated and real HSI datasets. Our method is compared with ten classical or state-of-the-art methods. Among them, four are based on low-rank matrix factorization: LRMR [17], LRTV [19], double low-rank (DLR) matrix decomposition [55], and correlated TV (CTV) [56]. Two methods utilize low-rank tensor decomposition: LRTDTV [24] and LRTDGS [25]. In addition, three subspace representation-based methods are included: LRTF-DFR [33], RCTV [34], and factor group sparsity-regularized nonconvex low-rank approximation (FGSLR) [57]. Finally, one deep learning-based method, FastHyMix [41], is also considered. The parameter settings for these comparison methods are adjusted according to the suggestions provided in the literature. To comprehensively evaluate the performance of different denoising methods, we analyze and compare them in terms of objective indexes and visual performance. Before denoising, the gray-scale values of the HSI are normalized to [0, 1] band-by-band. All methods are executed on MATLAB R2021a (Intel Core i9-11900 CPU at 2.50 GHz with 128 GB RAM).

A. Experimental Configurations

1) Experimental Data: Our experiments are implemented on two simulated noisy datasets and three real noisy datasets. The description of the datasets is given as follows.

1) *ROSIS Pavia City Center Dataset (PaviaC[1]):* This dataset is obtained by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor. The dataset has 102 bands in the spectral dimension. The first 22 bands are heavily affected by noise, and therefore, they cannot be used as references for noise removal experiments. To facilitate the experiment, we select a subimage with a size of 200 × 200 × 80.
2) *HYDICE Washington DC Mall Dataset (WDC[2]):* This dataset is obtained by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor. The dataset is 1208 × 307 pixels and has 191 spectral bands. In the experiment, a subimage sized 256×256×191 is selected.
3) *Gaofen 5 (GF-5) Shanghai and Baoqing[3]:* These two datasets are HSIs captured by the GF-5 satellite in Shanghai and Baoqing, China. Both of them are corrupted by Gaussian noise, dense dead lines, and stripes. The image size used in the experiment is 300×300×155.
4) *AVIRIS Indian Pines dataset[1]:* The Indian Pines data were captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in Northwestern Indiana, and the size is 145 × 145 × 200. Some bands in this dataset are seriously degraded by mixed Gaussian and impulse noises.

2) Experimental Indicators and Settings: To quantitatively evaluate the quality of the denoising performance of different methods, we employed three common indicators, i.e., the mean peak signal-to-noise ratio (MPSNR) [58], [59], the mean structural similarity index measure (MSSIM) [60], and the mean spectral angle mapper (MSAM) [61]. The larger values of MPSNR and MSSIM, coupled with lower values of MSAM, indicate better denoising results. The mathematical formulas for these indicators are given as follows:

$$\begin{aligned} \mathrm{MPSNR} &= \frac{1}{P}\sum_{b=1}^{P} 10\log_{10}\frac{{I_{\max,b}}^2 \times MN}{\sum_{x=1}^{M}\sum_{y=1}^{N}\left[\hat{u}_b(x,y) - u_b(x,y)\right]^2} \\ \mathrm{MSSIM} &= \frac{1}{P}\sum_{b=1}^{P}\frac{\left(2\mu_{u_b}\mu_{\hat{u_b}} + C_1\right)\left(2\sigma_{u_b\hat{u_b}} + C_2\right)}{\left(\mu_{u_b}^2 + \mu_{\hat{u_b}}^2 + C_1\right)\left(\sigma_{u_b}^2 + \sigma_{\hat{u_b}}^2 + C_2\right)} \\ \mathrm{MSAM} &= \frac{1}{P}\sum_{b=1}^{P}\arccos\left[\frac{\sum_i^{MN} u_b^i \hat{u}_b^i}{\sqrt{\sum_{i=1}^{MN}\left(u_b^i\right)^2}\sqrt{\sum_{i=1}^{MN}\left(\hat{u}_b^i\right)^2}}\right] \end{aligned} \tag{24}$$

where $I_{\max,b}$ denotes the maximum value of the bth band. u_b and $\hat{u}_b$ represent the clean image and the denoised image of the bth band, respectively. μ_{u_b} and $\mu_{\hat{u}_b}$ are the average values of clean HSI u_b and denoised HSI $\hat{u}_b$, respectively, while σ_{u_b} and $\sigma_{\hat{u}_b}$ denote the variances. $\sigma_{u_b\hat{u}_b}$ represents the covariance

[1]https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes
[2]https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html
[3]http://hipag.whu.edu.cn/resourcesdownload.html

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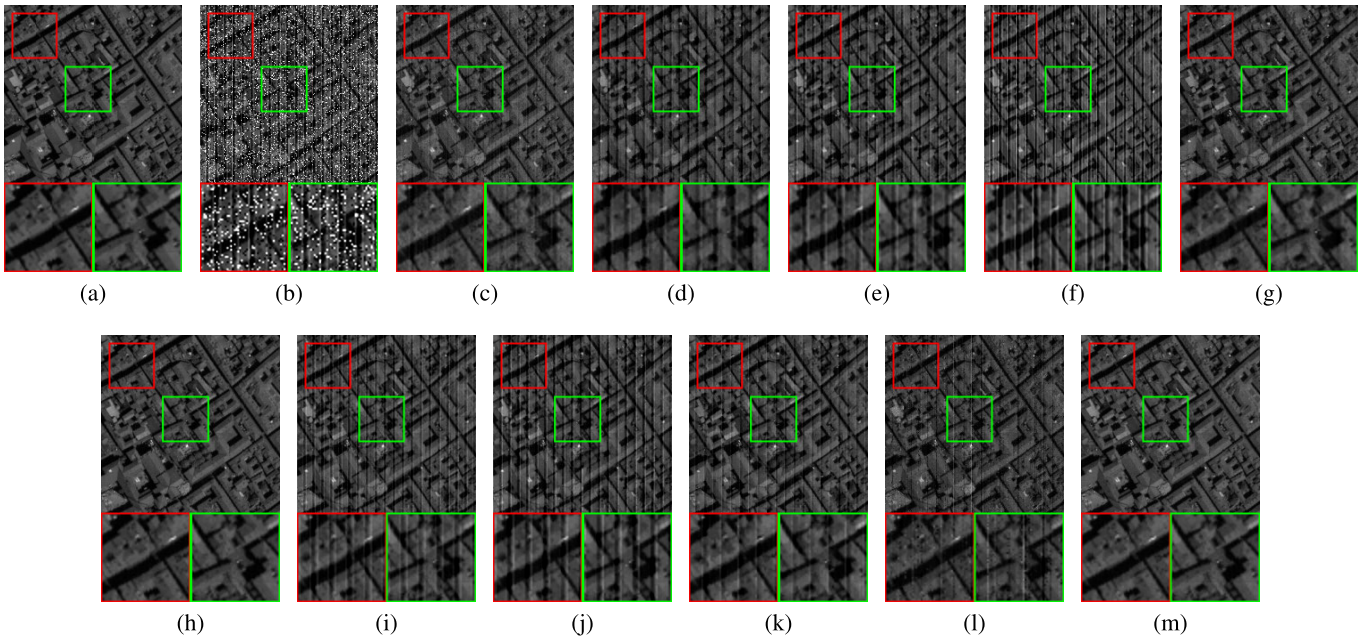


Fig. 6. Band 53 of the PaviaC dataset before and after denoising via the different methods under Case7. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) LRTDGS. (g) LRTF-DFR. (h) DLR. (i) CTV. (j) FGSLR. (k) RCTV. (l) FastHyMix. (m) HLTVSG.

between u_b and $\hat{u}_b$. M and N are the height and width of the spatial region, respectively, and P is the number of bands in the spectrum. In addition, C_1 and C_2 are very small positive constants.

In both simulated and real HSI experiments, the stopping criterion ξ is set to 10^{-4}, and the maximum number of iterations, $k_{\max}$, is set to 50. Furthermore, the upper bound rank r_s is set to 1, as recommended in [55]. The selection of the remaining important parameters will be discussed in Section IV-E1.

3) Simulation Configurations: We simulate the noisy HSI by adding a mixture of Gaussian noise, impulse noise, dead lines, and stripes to noise-free HSI. The specific descriptions are given as follows.

Case1: For each band, Gaussian noise with standard deviation $\sigma = 0.05$ and impulse noise with ratio $o = 0.1$ were added. Besides, 40% of the bands were randomly selected to add vertical random stripes, with the number of stripes being 30% of the image width and an intensity of 0.1.

Case2: Gaussian and impulse noises were added in the same manner as Case1. Furthermore, vertical periodic stripes were added in the same way as the vertical random stripes in Case1.

Case3: In this case, Gaussian noise with standard deviation $\sigma = 0 \sim 0.15$ and impulse noise with percentage $o = 0 \sim 0.2$ were added to each band. In addition, 30% of the bands were randomly selected to add vertical periodic stripes, with the number of stripes ranging from 30% to 50% of the image width and an intensity $0.1 \sim 0.2$. Furthermore, 10% of the bands were randomly selected to add dead lines, with the number of the dead lines ranging from 2 to 5 and the width of each dead line ranging from 1 to 3.

Case4: Gaussian noise, impulse noise, and dead lines were added as in Case3. In addition, 50% of the bands were randomly selected to add vertical periodic stripes, with the number of stripes ranging from 30% to 50% of the image width and an intensity $0.1 \sim 0.2$.

Case5: Gaussian noise, impulse noise, and dead lines were added in the same manner as in Case3. In addition, vertical random stripes were added in the same way as the vertical periodic stripes in Case3.

Case6: Gaussian noise, impulse noise, and dead lines were added in the same manner as in Case3. In addition, vertical random stripes were added in the same way as the vertical periodic stripes in Case4.

Case7: Gaussian noise, impulse noise, and dead lines were added as in Case3. Furthermore, a mixture of vertical random and periodic stripes was added. Specifically, 20% of the bands were randomly selected to add random stripes, with the number ranging from 20% to 40% of the image width and an intensity $0.1 \sim 0.2$. Another 20% of the bands were randomly selected to add periodic stripes with similar specifications.

Case8: Gaussian noise, impulse noise, and dead lines were added as in Case3. A mixture of vertical random and periodic stripes was added, with 40% of the bands randomly selected for each type of stripe, with the number ranging from 20% to 40% of the image width and an intensity $0.1 \sim 0.2$.

B. Simulated Datasets' Experiments

1) Visual Comparison: In terms of visual quality, we selected two representative datasets, PaviaC and WDC, both under Case7. Figs. 6 and 7 display the denoising results of different methods for the 53rd and 81st bands of the PaviaC and WDC datasets, respectively. For better visual comparison, we have zoomed in on certain regions of the images. The results indicate that all methods can remove the noise to varying degrees. Specifically, LRMR, LRTV, LRTDTV, LRTDGS, LRTF-DFR, CTV, FGSLR, RCTV, and FastHyMix

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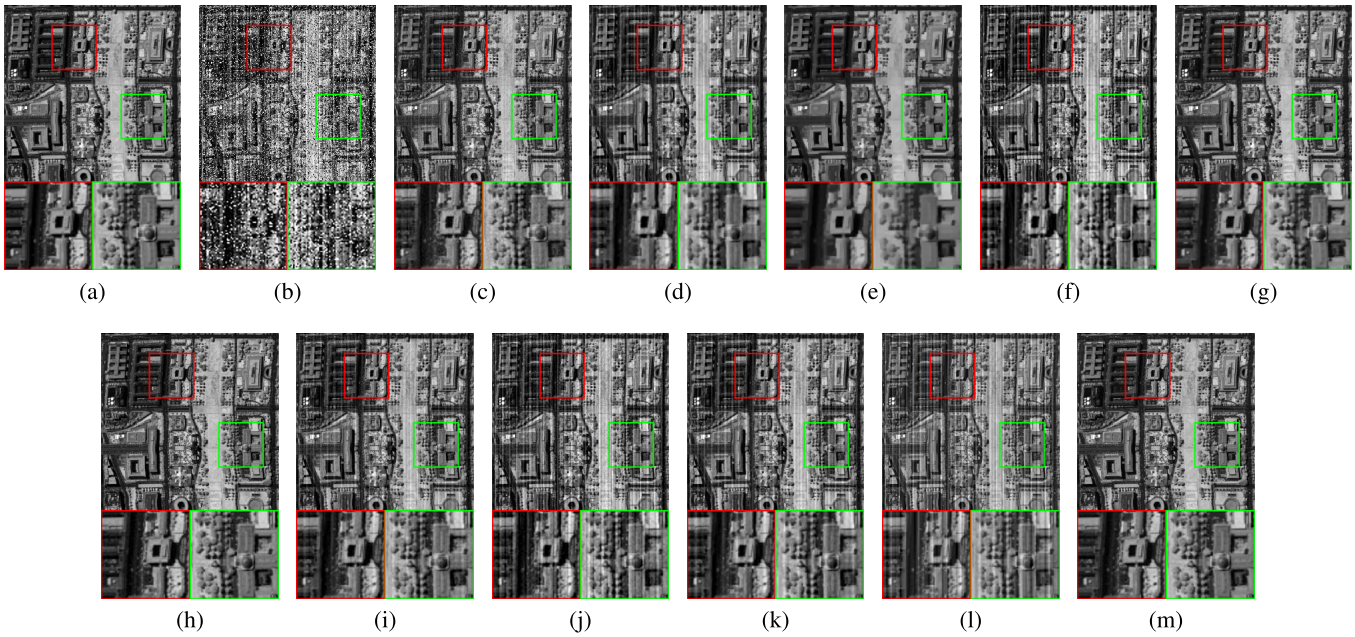


Fig. 7. Band 81 of the WDC dataset before and after denoising via the different methods under Case7. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) LRTDGS. (g) LRTF-DFR. (h) DLR. (i) CTV. (j) FGSLR. (k) RCTV. (l) FastHyMix. (m) HLTVSG.

TABLE I

QUANTITATIVE INDICES OF THE PAVIAC DATASET WITH DIFFERENT METHODS UNDER DIFFERENT CASES OF SIMULATED MIXED NOISE

Case	Indicators	Noisy	LRMR	LRTV	LRTDTV	LRTDGS	LRTF_DFR	DLR	CTV	FGSLR	RCTV	FastHyMix	HLTVSG
Case1	MPSNR	14.1548	33.4449	35.4749	36.0032	34.9758	36.4254	35.1180	35.5261	37.3512	37.2813	37.3205	**37.4197**
	MSSIM	0.2373	0.9404	0.9574	0.9623	0.9514	0.9671	0.9566	0.9598	0.9726	0.9716	**0.9753**	0.9684
	MSAM	0.7127	0.1186	0.0828	0.0833	0.1018	0.0727	0.1048	0.1093	**0.0689**	0.0782	0.0695	0.0855
Case2	MPSNR	14.1592	33.3656	34.5686	35.2347	32.5026	35.4964	34.9906	34.1490	34.8446	36.1341	34.9364	**37.4858**
	MSSIM	0.2358	0.9365	0.9438	0.9531	0.9086	0.9664	0.9564	0.9308	0.9336	0.9552	0.9585	**0.9688**
	MSAM	0.7223	0.1271	0.1107	0.1033	0.1661	0.0851	0.1072	0.1622	0.1445	0.1082	0.1043	**0.0836**
Case3	MPSNR	14.2778	30.5707	32.9661	33.9635	30.7914	31.1269	34.3521	32.8579	33.7267	32.2644	34.0898	**36.7873**
	MSSIM	0.2403	0.8836	0.9122	0.9282	0.8562	0.8940	0.9519	0.8891	0.8936	0.8943	0.9463	**0.9634**
	MSAM	0.7535	0.1647	0.1404	0.1266	0.2392	0.1554	0.1164	0.2095	0.2004	0.1976	0.1364	**0.0883**
Case4	MPSNR	14.1038	29.3363	30.0562	31.0165	27.6577	29.5998	33.3181	30.0953	29.5296	28.9138	30.4426	**36.5213**
	MSSIM	0.2295	0.8426	0.8444	0.8627	0.7627	0.8526	0.9287	0.7907	0.7403	0.8075	0.8549	**0.9625**
	MSAM	0.7680	0.2024	0.2373	0.2152	0.3120	0.2112	0.2497	0.3035	0.3661	0.2741	0.2529	**0.0907**
Case5	MPSNR	14.2047	30.7231	34.0083	34.9128	33.4773	33.1750	34.4261	33.8422	36.6492	35.7342	34.8733	**36.7577**
	MSSIM	0.2369	0.8953	0.9378	0.9467	0.9236	0.9299	0.9519	0.9308	**0.9688**	0.9544	0.9643	0.9632
	MSAM	0.7473	0.1514	0.1312	0.1070	0.1394	0.1081	0.1149	0.1484	**0.0747**	0.1067	0.1250	0.0868
Case6	MPSNR	13.8542	29.7712	33.4318	34.5731	33.1033	32.8809	34.2995	32.0488	36.0469	35.2155	33.6356	**36.6347**
	MSSIM	0.2200	0.8804	0.9319	0.9473	0.9245	0.9267	0.9520	0.9083	**0.9651**	0.9518	0.9579	0.9630
	MSAM	0.7579	0.1703	0.1271	0.1082	0.1407	0.1112	0.1150	0.1762	**0.0774**	0.1102	0.1297	0.0878
Case7	MPSNR	14.1304	30.7303	33.2437	34.3640	31.6536	32.4151	34.2773	33.1864	35.4496	33.6777	32.0512	**36.6527**
	MSSIM	0.2334	0.8949	0.9215	0.9355	0.8817	0.9206	0.9508	0.9142	0.9442	0.9232	0.9278	**0.9625**
	MSAM	0.7527	0.1541	0.1322	0.1256	0.2045	0.1295	0.1182	0.1730	0.1229	0.1478	0.1676	**0.0907**
Case8	MPSNR	13.9247	29.6649	32.2797	33.6376	30.0498	31.9058	33.9251	31.0130	32.1265	32.0924	30.7922	**36.4601**
	MSSIM	0.2234	0.8677	0.9116	0.9318	0.8426	0.9107	0.9500	0.8644	0.8683	0.8973	0.9102	**0.9622**
	MSAM	0.7695	0.1824	0.1361	0.1226	0.2359	0.1344	0.1208	0.2203	0.2195	0.1749	0.1930	**0.0899**

fail to fully eliminate mixed noise, particularly stripe noise, as they simply treat it as sparse noise without considering its unique characteristics. Consequently, these methods struggle to effectively remove stripe noise when it is nonsparse or periodically distributed. In contrast, both DLR and our proposed method explicitly model stripe noise separately, resulting in near-complete noise removal. However, DLR has difficulty restoring local details, particularly edges, as evident in the zoomed-in regions of Figs. 6 and 7. In comparison, our proposed method not only completely removes noise but also effectively preserves both the global structure and local details of the original HSI.

2) Quantitative Comparison: Tables I and II present the quantitative results of MPSNR, MSSIM, and MSAM for different methods under various noise conditions. The optimal values are highlighted in bold, and the suboptimal values are underlined. In most cases, the proposed method ranks first or second in terms of MPSNR, MSSIM, and MSAM. Although FGSLR shows better results in certain cases, its performance is unstable. For instance, in Table II, Cases 4 and 7 fail to provide satisfactory results. In contrast, the proposed method demonstrates a more stable performance.

3) Mean Vertical Profiles: Figs. 8 and 9 show the mean vertical profiles for the 53rd band of the PaviaC dataset and

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TABLE II

QUANTITATIVE INDICES OF THE WDC DATASET WITH DIFFERENT METHODS UNDER DIFFERENT CASES OF SIMULATED MIXED NOISE

Case	Indicators	Noisy	LRMR	LRTV	LRTDTV	LRTDGS	LRTF-DFR	DLR	CTV	FGSLR	RCTV	FastHyMix	HLTVSG
Case1	MPSNR	14.1237	33.7696	34.1490	30.5853	36.0272	36.8031	26.6579	36.0486	**38.5844**	37.7796	37.1043	37.6369
	MSSIM	0.2127	0.8916	0.8817	0.7568	0.9234	0.9459	0.8710	0.9551	0.9624	0.9499	**0.9687**	0.9677
	MSAM	0.6697	0.0981	0.0636	0.0882	0.0699	0.0571	0.3127	0.0738	**0.0483**	0.0573	0.0639	0.0577
Case2	MPSNR	14.1585	33.4003	32.1001	30.5216	32.4311	35.8493	26.9040	33.8747	33.8048	33.9790	33.2095	**36.8106**
	MSSIM	0.2127	0.8750	0.8363	0.7630	0.8352	0.9415	0.8733	0.8936	0.8315	0.8772	0.8973	**0.9657**
	MSAM	0.6749	0.1041	0.1058	0.0878	0.1366	**0.0618**	0.3159	0.1277	0.1582	0.1117	0.1270	0.0636
Case3	MPSNR	14.0947	30.1054	30.4628	29.2295	30.5698	32.9179	27.4775	32.5175	33.0491	32.5164	30.3760	**35.7444**
	MSSIM	0.2008	0.7899	0.7798	0.6991	0.7812	0.8729	0.8710	0.8502	0.7943	0.8304	0.8414	**0.9518**
	MSAM	0.7047	0.1657	0.1493	0.1004	0.2195	0.1119	0.2916	0.1784	0.2380	0.1666	0.1973	**0.0717**
Case4	MPSNR	13.9022	28.1400	28.2110	28.6333	28.0495	29.7266	27.4786	29.2158	29.4390	29.1154	27.6076	**35.3640**
	MSSIM	0.1888	0.7088	0.6899	0.6792	0.6818	0.7588	0.8674	0.7408	0.6754	0.7182	0.7383	**0.9510**
	MSAM	0.7175	0.2115	0.2004	0.1093	0.2609	0.1779	0.2806	0.2535	0.3020	0.2332	0.2553	**0.0752**
Case5	MPSNR	14.1064	30.7121	32.1119	29.2664	34.0181	34.4564	27.6000	34.0340	**36.8969**	35.5784	33.0742	36.1672
	MSSIM	0.2035	0.8261	0.8312	0.6947	0.8924	0.9192	0.8685	0.9272	0.9326	0.9180	0.9431	**0.9535**
	MSAM	0.7014	0.1459	0.0915	0.0990	0.1163	0.0833	0.2705	0.1024	0.1570	0.0869	0.1302	**0.0665**
Case6	MPSNR	13.8639	29.2825	31.3329	28.8382	32.7710	34.1193	27.3341	32.0490	33.6303	34.3343	33.0742	**36.1445**
	MSSIM	0.1902	0.7810	0.8081	0.6712	0.8584	0.9145	0.8656	0.8886	0.8636	0.9003	0.9111	**0.9540**
	MSAM	0.7110	0.1695	0.0965	0.1045	0.1173	0.0855	0.2734	0.1291	0.1309	0.0926	0.1434	**0.0668**
Case7	MPSNR	14.1017	30.5495	31.0132	29.2715	32.6597	34.3026	27.6242	33.1844	34.3625	33.8117	31.4330	**35.9505**
	MSSIM	0.2014	0.8121	0.8027	0.6964	0.8414	0.9150	0.8674	0.8940	0.8484	0.8714	0.8823	**0.9530**
	MSAM	0.7042	0.1473	0.1216	0.0979	0.1551	0.0842	0.2650	0.1289	0.1725	0.1256	0.1664	**0.0697**
Case8	MPSNR	13.7441	28.3031	29.5611	28.6955	28.9989	32.2350	26.9923	30.0155	30.4514	31.1357	28.7404	**35.4853**
	MSSIM	0.1819	0.7330	0.7450	0.6678	0.7348	0.8574	0.8666	0.8107	0.7387	0.7965	0.8000	**0.9514**
	MSAM	0.7211	0.1959	0.1551	0.1060	0.2262	0.1185	0.2975	0.1918	0.2431	0.1754	0.2126	**0.0747**

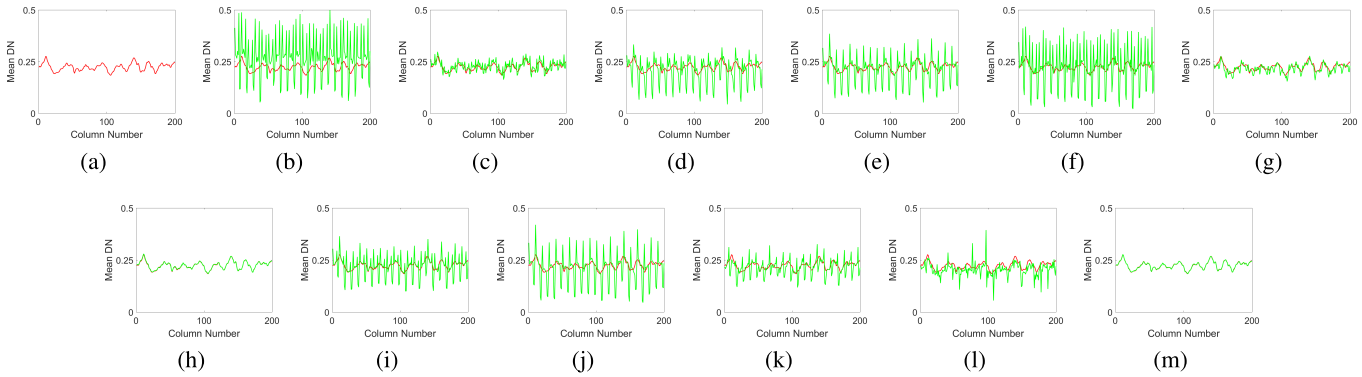


Fig. 8. Mean vertical profiles of band 53 in the PaviaC dataset (Case7). (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) LRTDGS. (g) LRTF-DFR. (h) DLR. (i) CTV. (j) FGSLR. (k) RCTV. (l) FastHyMix. (m) HLTVSG.

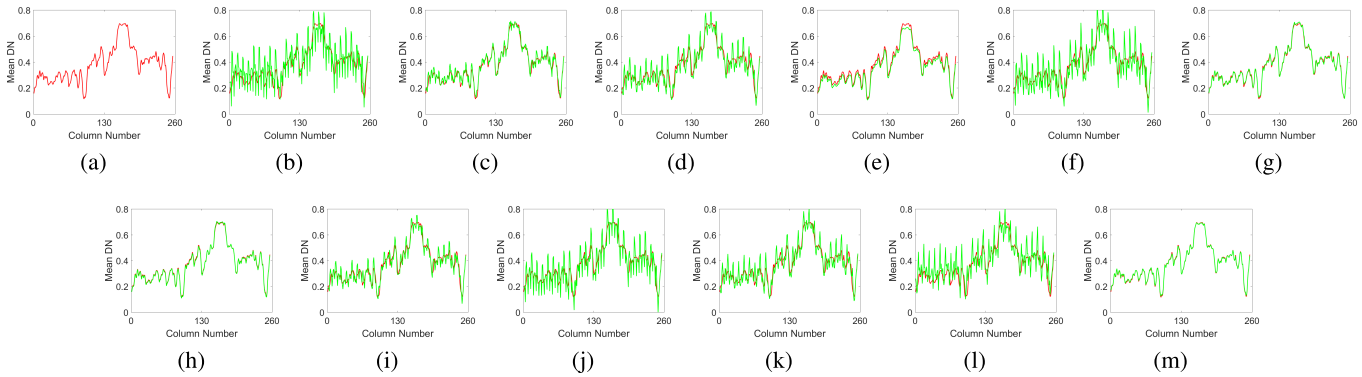


Fig. 9. Mean vertical profiles of band 81 in the WDC dataset (Case7). (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) LRTDGS. (g) LRTF-DFR. (h) DLR. (i) CTV. (j) FGSLR. (k) RCTV. (l) FastHyMix. (m) HLTVSG.

the 81st band of the WDC dataset, respectively. In these figures, the horizontal axis represents the column index, and the vertical axis represents the mean digital number value of each column. The red curve represents the mean vertical profile of the clean HSI, while the green curve denotes the noisy HSI or the image denoised by different methods. It is evident that the curves for LRMR, LRTV, LRTDTV, LRTDGS, CTV, FGSLR, RCTV, and FastHyMix are severely degraded due to the presence of stripe noise, whereas the curves for LRTF-DFR, DLR, and the proposed method are closer to the

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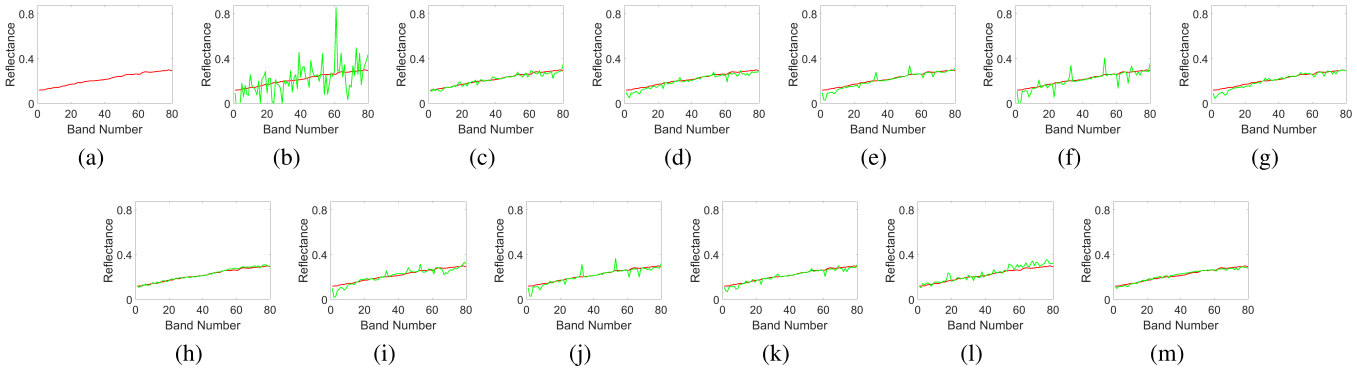


Fig. 10. Original spectrum (red curve) and the denoising results (green curve) of pixel (61, 162) in simulated Case7 of the PaviaC dataset. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) LRTDGS. (g) LRTF-DFR. (h) DLR. (i) CTV. (j) FGSLR. (k) RCTV. (l) FastHyMix. (m) HLTVSG.

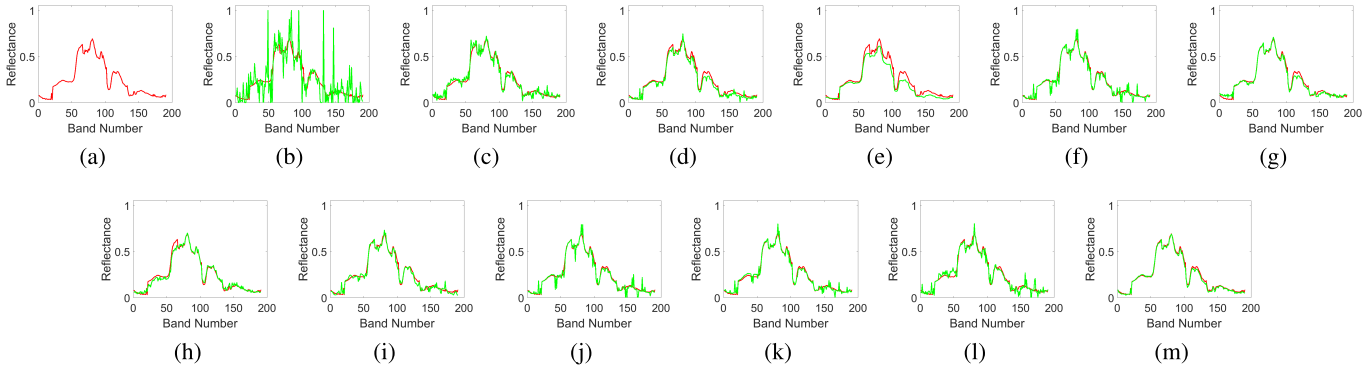


Fig. 11. Original spectrum (red curve) and the denoising results (green curve) of pixel (176, 71) in simulated Case7 of the WDC dataset. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) LRTDGS. (g) LRTF-DFR. (h) DLR. (i) CTV. (j) FGSLR. (k) RCTV. (l) FastHyMix. (m) HLTVSG.

clean HSI. Moreover, a careful comparison reveals that the mean vertical profiles obtained by our proposed method are closer to the clean HSI across both datasets.

4) Spectrum Curves: To analyze the spectral recovery capability of each method, we compared the spectral curves. Figs. 10 and 11 show the original spectra (red curve) and denoised spectra (green curve) for pixels (61, 162) of the PaviaC dataset and (176, 71) of the WDC dataset, respectively. It is evident that the spectral curves recovered by the proposed method are most consistent with the original spectra, while others are more or less impacted by noise. These curves further demonstrate the superiority and robustness of the proposed method.

5) Overall PSNR and SSIM Curves of Each Band: Fig. 12 shows the PSNR and SSIM curves for different methods applied to the PaviaC dataset Case7 and the WDC dataset Case8. As illustrated in the figure, the PSNR and SSIM curves for most methods exhibit noticeable sawtooth patterns, particularly in bands affected by stripe noise, which is present in a few randomly selected bands. The proposed method treats stripe noise as an independent component, effectively mitigating its impact and producing relatively smoother curves.

For both datasets, the proposed method (red curve) consistently delivers higher and more stable PSNR and SSIM values across most bands compared to other methods. Although the PSNR curve of FGSLR exceeds that of the proposed method in some bands, its overall MPSNR is lower due to its inability to fully eliminate stripe noise. In addition, the proposed method offers a significant advantage in runtime compared to FGSLR. These results further demonstrate the reliability and effectiveness of the proposed method.

C. Real Noisy Datasets' Experiments

Fig. 13 shows the pseudocolor images of the Baoqing dataset, composed of the (155, 152, 86) bands, along with the corresponding denoising results of different methods. It can be observed that the original image is affected by mixed noise, particularly stripe noise. After denoising, methods such as LRMR, LRTV, LRTDTV, and LRTDGS still exhibit obvious stripe noise. In contrast, LRTF-DFR, CTV, FGSLR, and RCTV demonstrate better visual performance but inevitably produce stripe artifacts. Furthermore, FastHyMix, DLR, and the proposed method provide the best visual results. Fig. 14 displays the pseudocolor image of the Shanghai dataset, composed of the (38, 152, 155) bands. The original pseudocolor image is heavily contaminated by a mixture of Gaussian noise, sparse noise, and stripe noise. In this dataset, the denoising results of LRMR, LRTV, LRTDTV, LRTDGS, CTV, FGSLR, and RCTV are unsatisfactory. LRTF-DFR and FastHyMix offer slightly improved visual quality but still present some artifacts. Both DLR and the proposed method exhibit the best denoising performance while preserving the global structure of the image. In addition, Fig. 15 shows the pseudocolor images of the Indian Pines dataset, composed of the (108, 140, 220) bands. All methods demonstrate varying degrees of denoising effectiveness on the noisy bands. However, LRMR

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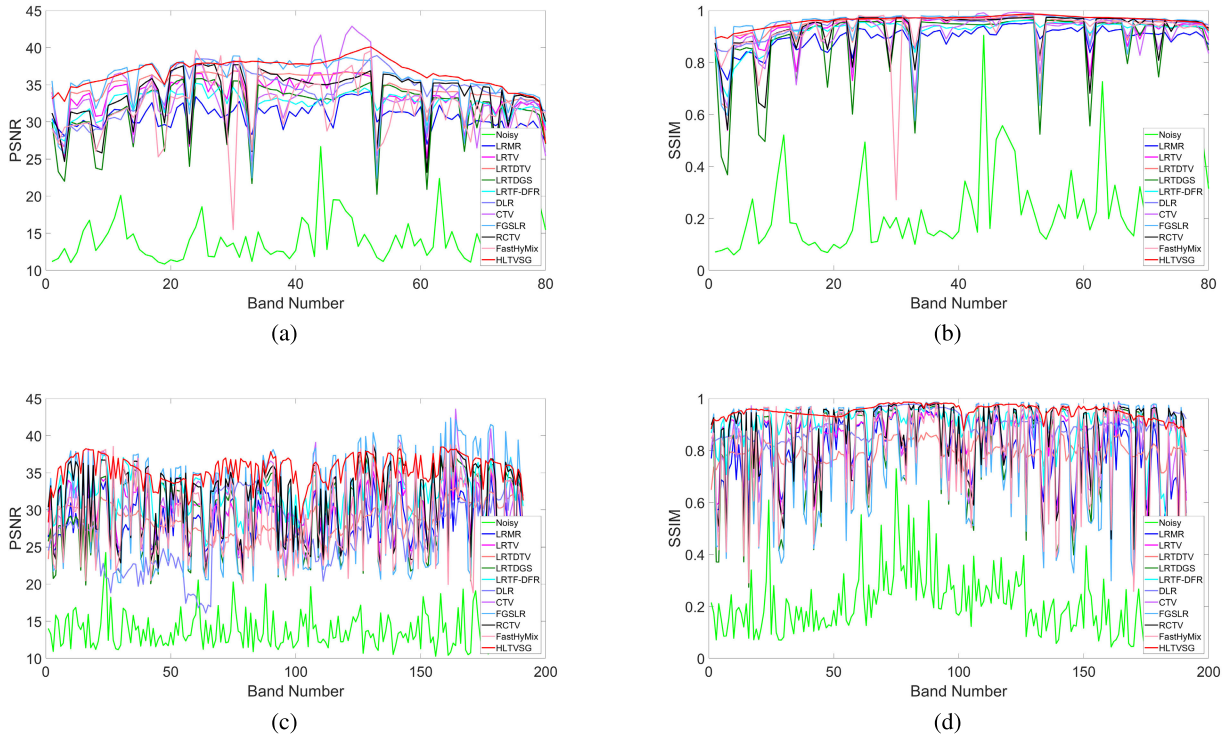


Fig. 12. Overall PSNR and SSIM values of the 11 competing methods. (a) and (b) Indexes of the PaviaC dataset (Case7). (c) and (d) Indexes of the WDC dataset (Case8).

TABLE III
RUNNING TIME (Seconds) OF REAL DATA EXPERIMENTS

Data	Size	LRMR	LRTV	LRTDTV	LRTDGS	LRTF-DFR	DLR	CTV	FGSLR	RCTV	FastHyMix	HLTVSG
Baoqing	$300 \times 300 \times 155$	48.3	50.8	126.6	94.0	74.4	28.3	69.0	64.8	9.6	11.1	57.3
Shanghai	$300 \times 300 \times 155$	43.7	48.6	130.8	87.5	66.1	30.0	75.0	58.4	9.3	8.4	50.2
Indian Pines	$145 \times 145 \times 220$	20.0	19.4	87.9	59.5	42.8	31.8	33.4	67.0	4.8	4.4	35.2

and DLR show noticeable noise residues. Methods such as LRTV, LRTDTV, LRTDGS, and FGSLR effectively remove the noise but smooth the structure to varying degrees. LRTF-DFR, CTV, RCTV, and HLTVSG achieve reliable results, with HLTVSG better preserving fine spatial structures, as observed in the zoomed-in region. Furthermore, Table III lists the running time of different methods applied to the real noisy datasets. While the running time of the proposed method is not the shortest in comparison, it remains competitive with other methods. Considering both denoising performance and computational cost, it can be seen that the proposed method comprehensively outperforms the other methods.

D. Classification Results

To further analyze the restored HSIs using different denoising methods, we conduct HSI classification experiments on the real Indian Pines dataset. The support vector machine (SVM) is employed to classify all recovery results of Indian Pines under the same conditions. Overall accuracy (OA) and the Kappa coefficient are used as evaluation metrics to quantitatively assess the classification performance. We perform PCA for dimensionality reduction, retaining the first 30 principal components. Then, we randomly sample 30% of the training samples, with the remaining samples used as the testing set for each class. In this study, 16 land-cover classes are included to validate the classification results. The classification outcomes are visualized in Fig. 16. In addition, the OA and Kappa coefficient values are shown in Table IV. As indicated in the table, the overall classification performance generally improves after applying the HSI denoising methods, demonstrating that the denoising process for HSIs is effective and significant. Moreover, the proposed HLTVSG method achieves the highest OA and Kappa coefficient, highlighting the superiority of the proposed HSI denoising method.

E. Discussion

1) Parameter Analysis: The method proposed in this article involves three important parameters: rank r and regularization parameters λ and β. The selection of these parameters directly affects the denoising performance of HSI. In this section, we explore the optimal settings for these parameters through experiments. Moreover, the simulated datasets Case1, Case2, Case3, Case5, and Case7, representing different experimental conditions, are selected for analysis.

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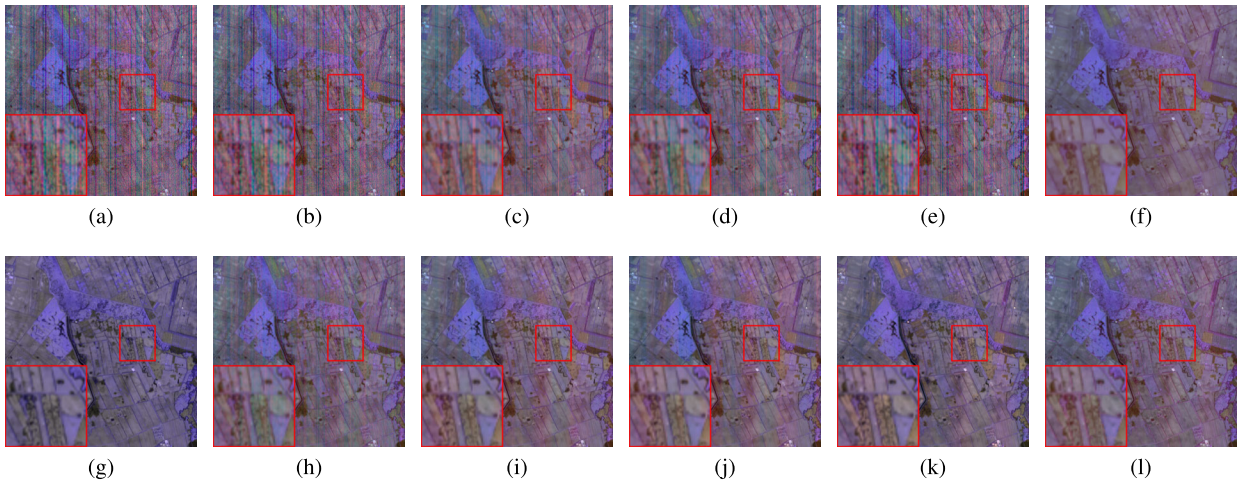


Fig. 13. Restoration results for the GF-5 Baoqing image in bands (155, 152, 86) via the different methods. (a) Original image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) LRTDGS. (f) LRTF-DFR. (g) DLR. (h) CTV. (i) FGSLR. (j) RCTV. (k) FastHyMix. (l) HLTVSG.

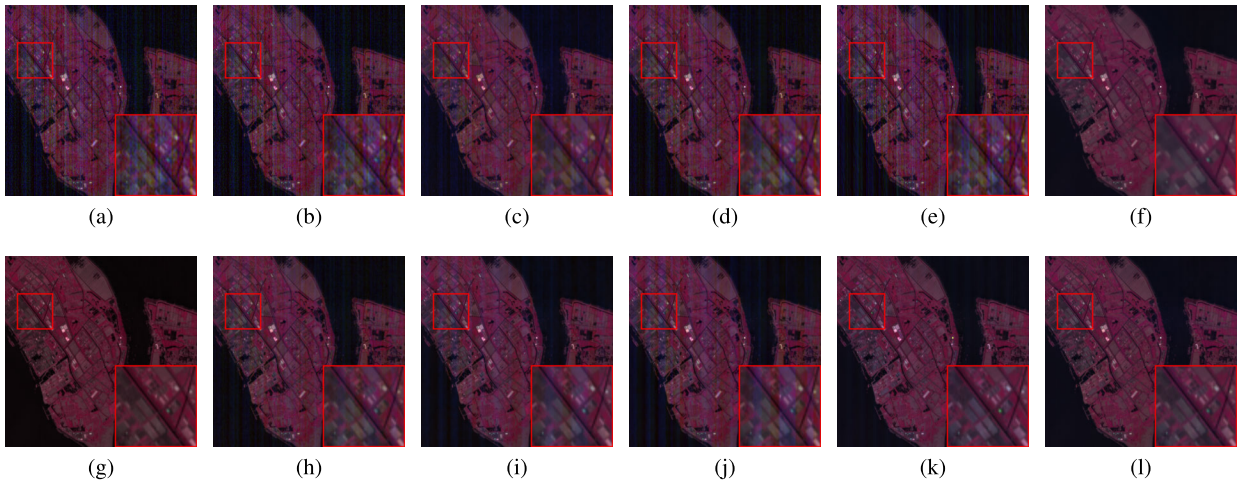


Fig. 14. Restoration results for the GF-5 Shanghai image in bands (38, 152, 155) via the different methods. (a) Original image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) LRTDGS. (f) LRTF-DFR. (g) DLR. (h) CTV. (i) FGSLR. (j) RCTV. (k) FastHyMix. (l) HLTVSG.

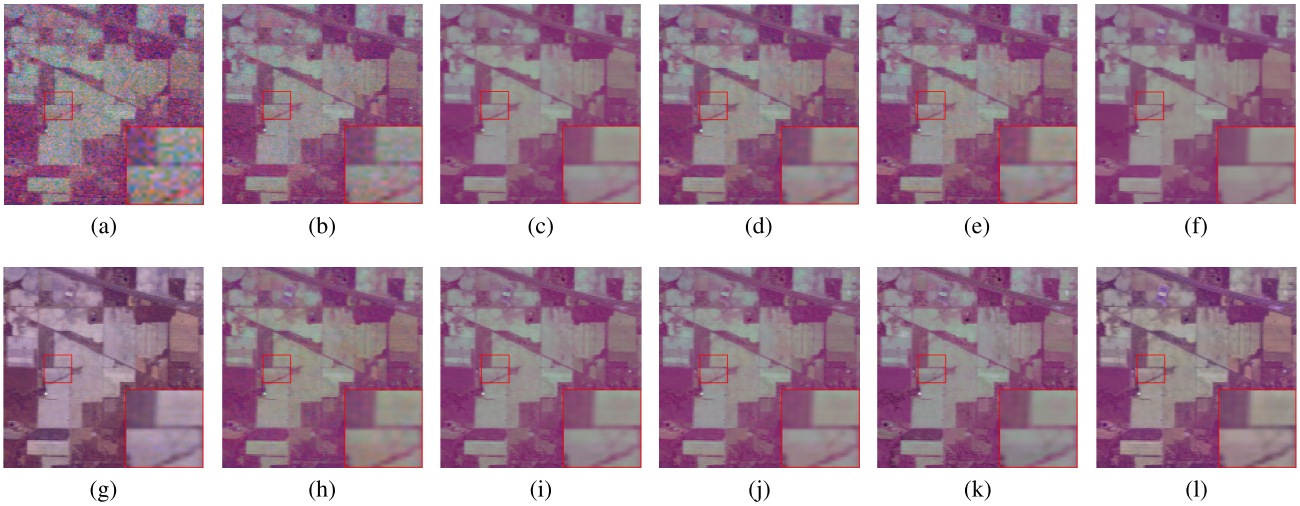


Fig. 15. Restoration results for the Indian Pines image in bands (108, 140, 220) via the different methods. (a) Original image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) LRTDGS. (f) LRTF-DFR. (g) DLR. (h) CTV. (i) FGSLR. (j) RCTV. (k) FastHyMix. (l) HLTVSG.

The upper bound rank r mainly characterizes the global spectral correlation of HSI. The optimal subspace dimension may vary across different datasets. Experimental results indicate that the proposed method achieves the best denoising performance when the subspace dimensions are 4 and 2 for the WDC and PaviaC datasets, respectively, as shown in

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TABLE IV
CLASSIFICATION RESULT ON THE REAL INDIAN PINES DATASET

Index	Noisy	LRMR	LRTV	LRTDTV	LRTF-DFR	DLR	CTV	FGSLR	RCTV	FastHyMix	HLTVSG
OA	0.5689	0.7129	0.7377	0.7116	0.8927	0.7151	0.8584	0.7347	0.8086	0.7669	**0.8936**
Kappa	0.5451	0.6957	0.7218	0.6945	0.8860	0.6978	0.8496	0.7190	0.7969	0.7528	**0.8870**

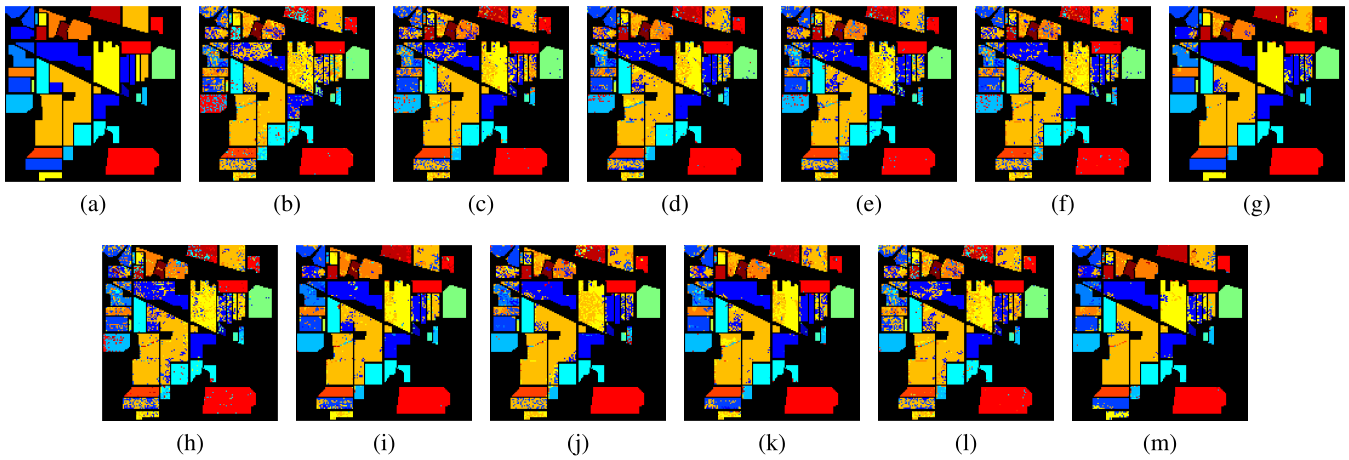


Fig. 16. HSI classification results for the Indian Pines data before and after HSI denoising. (a) Ground truth. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) LRTDGS. (g) LRTF-DFR. (h) DLR. (i) CTV. (j) FGSLR. (k) RCTV. (l) FastHyMix. (m) HLTVSG.

TABLE V
DIFFERENT VARIANT MODELS FOR ABLATION STUDY

Model 1	HLTVSG with $\lambda = 0$
Model 2	HLTVSG with $\beta = 0$
Model 3	HLTVSG with the first term replaced by $\|D_3(\mathcal{X})\|_* + \tau\|D_3(\mathcal{X})\|_{\text{HLTV}}$
Model 4	HLTVSG with the first term replaced by $\|\mathcal{U}\|_{\text{TV}}$

Fig. 17. Therefore, in our experiments, r is fixed at 2 for the PaviaC dataset and 4 for the WDC dataset. Furthermore, in the three real data experiments, the rank r is estimated using the HySime algorithm [28].

Parameters λ and β are employed to constrain the stripe noise term and the sparse noise term, respectively. Fig. 18 presents the sensitivity analysis of λ and β. The results show that the proposed algorithm is robust to variations in λ but sensitive to changes in β. The trend in MPSNR values across different noise conditions remains consistent as λ and β are varied, demonstrating the stability of these parameters under varying noise conditions. In addition, the best performance is achieved when λ is in the range [1.7, 3.8] and β is in the range [2, 4.7]. Therefore, we recommend selecting λ and β from the aforementioned ranges based on the degree of noise.

2) Ablation Experiment: The proposed HLTVSG model incorporates three key modules: the low-rank smoothness term $\sum_{i=1}^{2} \|\mathcal{Z}_i\|_{q_i}^{q_i}$, the low-rank term $\sum_{b=1}^{P} \|\mathbf{S}_b\|_*$, and the sparse term $\|\mathcal{B}\|_1$. To evaluate the effectiveness of these individual modules, we consider several variant models listed in Table V.

1) *Model 1:* HLTVSG without sparse term $\|\mathcal{B}\|_1$, i.e., with $\lambda = 0$, removing the sparse noise $\mathcal{B}$.
2) *Model 2:* HLTVSG without low-rank term $\sum_{b=1}^{P} \|\mathbf{S}_b\|_*$, i.e., with $\beta = 0$, removing the stripe noise $\mathcal{S}$.
3) *Model 3:* HLTVSG with the first term replaced by $\|D_3(\mathcal{X})\|_* + \tau\|D_3(\mathcal{X})\|_{\text{HLTV}}$. We apply both low-rank constraints and HLTV constraints directly to $D_3(\mathcal{X})$, i.e., the spectral gradient of the HSI. This experiment isolates the impact of subspace representation by demonstrating how it captures the low-rank structure inherent in the hyperspectral data.
4) *Model 4*, HLTVSG with the first term replaced by $\|\mathcal{U}\|_{\text{TV}}$. We apply traditional TV regularization to the subspace representation coefficients without incorporating the hyper-Laplacian distribution. This allows us to specifically evaluate the contribution of HLTV regularization in preserving spatial smoothness and its role in improving denoising performance.

As demonstrated by the results of the five models tested on the simulated PaviaC dataset (Case3), as presented in Table VI and Fig. 19, each module of HLTVSG plays an important role in effectively removing mixed noise. Model 1, which lacks the sparse noise constraint, performs the worst, as it struggles to handle sparse noise effectively. Model 2, which fails to adequately capture the characteristics of stripe noise, struggles to effectively disentangle it. As shown in Fig. 19, there remains a lot of stripes in the denoising result. Model 3, which skips subspace decomposition and applies denoising directly to the spectral gradients, performs better than Models 1 and 2, but it still does not fully capture the spectral correlations. Model 4, which uses traditional TV regularization instead of HLTV, offers some improvement but still falls short compared to our method. It is evident that satisfactory denoising results are achieved when all modules are correctly integrated.

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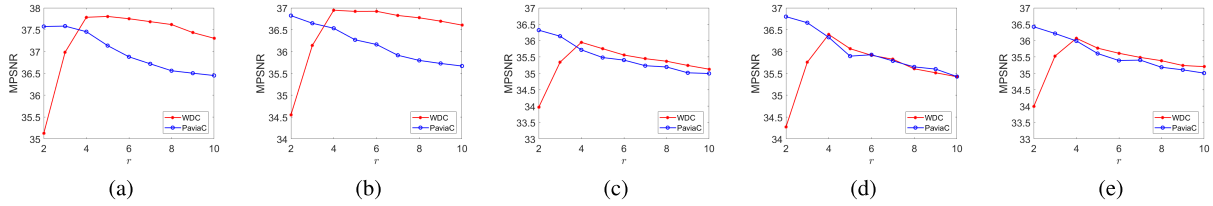


Fig. 17. Sensitivity analysis to parameter r. (a) Case1. (b) Case2. (c) Case3. (d) Case5. (e) Case7.

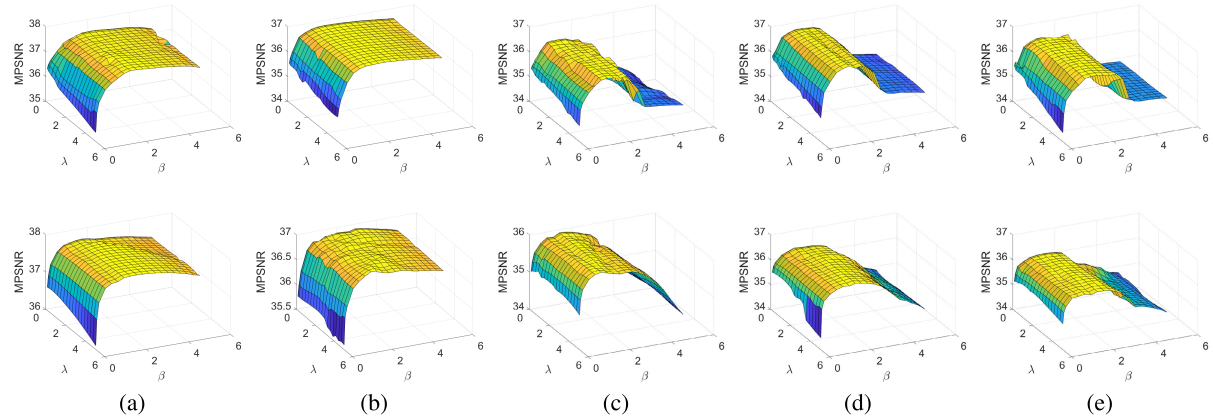


Fig. 18. Sensitivity analysis to parameters λ and β for the PaviaC dataset (Top) and the WDC dataset (Bottom). (a) Case1. (b) Case2. (c) Case3. (d) Case5. (e) Case7.

TABLE VI
QUANTITATIVE ANALYSIS OF HLTVSG ON THE PAVIAC DATASET CASE3

Model	MPSNR	MSSIM	MSAM
Model 1	26.0537	0.8016	0.1583
Model 2	33.4221	0.9194	0.1631
Model 3	36.0574	0.9533	0.1180
Model 4	36.4787	0.9617	0.0903
HLTVSG	**36.7873**	**0.9634**	**0.0883**

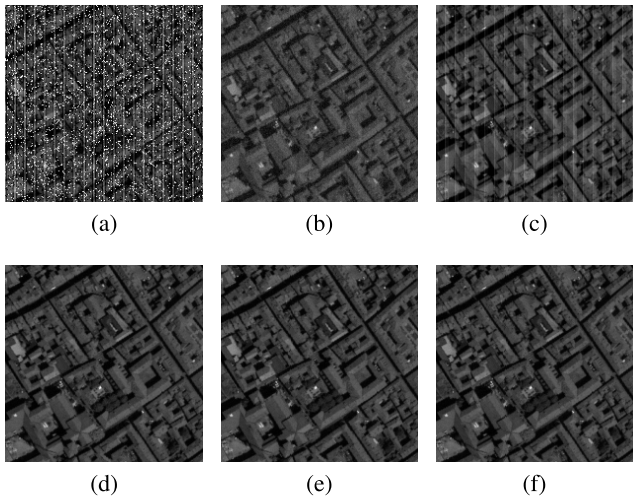


Fig. 19. Band 50 of the PaviaC dataset before and after denoising via the different models under Case3. (a) Noisy. (b) Model 1. (c) Model 2. (d) Model 3. (e) Model 4. (f) HLTVSG.

3) Power Parameter q Analysis: The power parameter q in HLTV term [see (5)] plays a crucial role in regularizing the smoothness. However, accurately estimating the optimal value of q under mixed noise conditions remains a challenging task [62]. To better understand the influence of q on denoising performance, we performed a case study under two specific datasets, PaviaC Case7 and WDC Case7, as shown in Fig. 20. The results from this case study provide valuable insights into the denoising effectiveness across different values of q, highlighting the tradeoffs between sparsity and noise reduction performance.

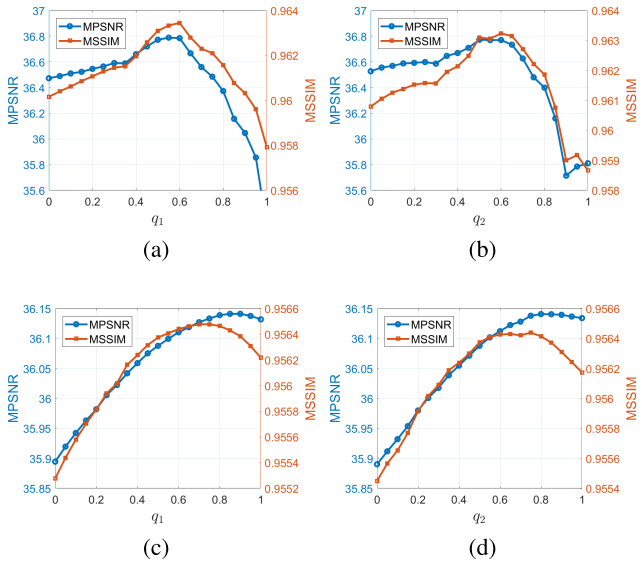


Fig. 20. Influence of q on denoising performance. (a) and (b) Results on PaviaC Case7. (c) and (d) Results on WDC Case7.

Drawing from the analysis conducted in several prior studies [48], [63] and the empirical results presented in this work,

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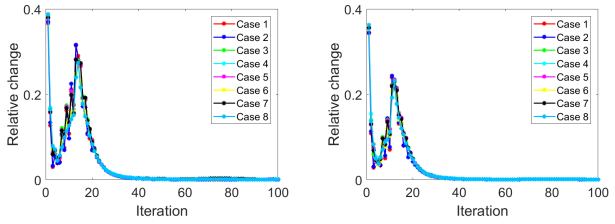


Fig. 21. Relative change curves with respect to the iteration number. (a) PaviaC dataset. (b) WDC dataset.

we recommend selecting q within the range [0.4, 0.7]. For the sake of consistency and simplicity, we set both q_1 and q_2 to 0.5 in our experiments.

4) Convergence Analysis: We conduct a numerical analysis of the convergence of the proposed method, employing the relative change $(\|\mathcal{X}^{k+1} - \mathcal{X}^k\|_F)/(\|\mathcal{X}^k\|_F)$ to empirically demonstrate the algorithm's convergence. Fig. 21 illustrates the relative change in the recovered HSI for each iteration of the PaviaC dataset and the WDC dataset under different noise cases compared to the previous iteration. The curves clearly show that as the number of iterations increases, the relative change initially experiences a sudden increase and then converges to zero, indicating that the proposed method is convergent.

5) Computational Complexity: The computational complexity of the proposed HLTVSG algorithm mainly consists of four parts.

1) Solution of $\mathcal{Z}$ and $\mathcal{B}$ subproblems via (9) and (18), which involves the soft-thresholding shrinkage and the GST operators. This requires $O(\text{MNP} + \text{MNrJ})$, where J is the number of iterations in the GST operator.
2) For (12) and (15), the computation involved in solving $\mathcal{X}$ and $\mathcal{U}$ depends on FFT and mainly costs $O(\text{MNP}\log(\text{MNP}) + \text{MNr}\log(\text{MNr}))$.
3) The solution of $\mathcal{S}$ through (20) has a complexity of $O(\text{MNPr}_s)$, where r_s is the upper bound rank of the stripe noise matrix.
4) Updating the Lagrange multiplier via (23), which requires $O(\text{MNP})$.

Therefore, the overall computational complexity of Algorithm is $O(k_{\max}(2\text{MNP} + \text{MNrJ} + \text{MNP}\log(\text{MNP}) + \text{MNr}\log(\text{MNr}) + \text{MNPr}_s))$, where $k_{\max}$ is the number of iterations.

V. Conclusion

In this article, an HSI denoising method via the HLTVSG domain has been proposed. We discovered that the low rankness and smoothness of HSIs can be more effectively leveraged within the spectral gradient domain. By capitalizing on this observation, we utilize a subspace representation approach to exploit the low-rank property of HSIs in the spectral gradient domain, which not only enhances denoising performance but also reduces computational costs. Furthermore, by examining the gradient distribution of SRCs, we found that using HLTV to constrain the spatial LSS of SRCs is more appropriate. Both simulation and real HSI data experiments were conducted, and the experimental results clearly demonstrate that compared with other classical and state-of-the-art denoising methods, the proposed method exhibits superior performance in both visual and quantitative evaluations.

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